

Shania Sinha

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EDUCATION

University of Amsterdam, Graduate School of Informatics Amsterdam, NL
MSc Computational Science Expected: July 2026

- Current GPA: 7.8
- Thesis: Social Influence in Climate-Driven Conflict (*Ongoing*)

Erasmus University Rotterdam, Erasmus School of Economics Rotterdam, NL
MSc Behavioral Economics August 2023

- GPA: 7.6
- Thesis: *E Pluribus Unum*: An Analysis of Conditions for Wisdom of Crowds

Purdue University, Krannert School Of Management West Lafayette, IN, USA
BSc Economics May 2022

- GPA: 3.33/4.0
- Minors: Mathematics, Computer Science

RESEARCH EXPERIENCE

Research Assistant January 2024 – September 2024
FutureTech Lab, MIT Cambridge, MA, USA

- Supervised by Dr. Neil Thompson and Dr. Sukrit Puri, conducting research on the economic impact of changes in semiconductor chip design. Conducting an event study to analyze the effect of the ‘end of Moore’s Law’ on firm productivity. Core data sanitation and analysis was conducted in R and Python.

Research Assistant May 2023 – July 2023
Erasmus School of Accounting & Assurance Rotterdam, NL

- Conducted meticulous dataset creation (including web scraping and data sanitation) on ICT project data sourced from the Dutch government, ensuring accuracy and integrity of the dataset.
- Further assisted in finding systematic indicators of planning issues in government-funded ICT projects.

Data Research Assistant August 2021 – May 2022
OHAIRE Lab, Purdue University West Lafayette, IN, USA

- Employed multi-level modeling techniques in R, SAS, and SPSS to examine the influence of full-time therapy dogs on the overall and momentary energy levels of pediatric healthcare professionals.
- Conducted dataest creation, data cleaning and exploratory data analysis of 31 large datasets on military veterans’ sleep, mood, and energy patterns using R to find the impact of service dogs on the recovery of veterans with Post-Traumatic Stress Disorder (PTSD).

Undergraduate Research Assistant August 2021 – Jan 2022
Krannert School of Management, Purdue University West Lafayette, IN, USA

- Built an interactive web-based multi-player strategy game in oTree for various ongoing research projects in the field of Experimental Economics. Implemented and designed the oTree web interface using JavaScript and HTML.

PUBLICATIONS & SELECTED RESEARCH PROJECTS

Individual Decision-Making in Coupled Agent-Environment Systems June-December, 2025
Link to project: [GitHub](#)

- Presenting and co-first author for **publication accepted to International Conference on Computational Science (CMAISS track)**, in Hamburg, Germany, in June 2026.
Co-authors: V. Peterson, K. Chlopicka, H. Zwart, Dr. D. Roy
- Built large-scale agent-based models to investigate how economic decision-making, social norms, and environmental feedback shape collective outcomes. Applied methods from Behavioural Economics and Environmental Science to explore emergent macro-patterns and phase transitions.
- Developed highly reproducible Python code for parameter sweeps and visualization, demonstrating capacity for interdisciplinary model design and empirical validation.

Modeling Refugee Displacement through Community Networks

June–August, 2025

Link to project: [GitHub](#)

- Project done under supervision of Dr. Diana Suleimenova and Dr. Derek Groen (Brunel University, London).
- Extended the *Flee3* agent-based displacement framework with community-network dynamics to study refugee flows, integrating insights from Political Science, Behavioural Economics, and Migration Studies. Designed new transition probability metrics to capture sequential refugee movements and validated results against empirical data from UNHCR, IOM, and ACLED.
- Focused on creating reproducible simulations and analysis pipelines, contributing to research relevant for humanitarian policy. Currently working towards an academic publication.

Economic Redistribution in Cellular Automata

January, 2025

Link to project: [GitHub](#)

- Implemented a cellular automata framework to analyze redistribution policies, linking macroeconomic theory (specifically wealth inequality and taxation) with computational modeling techniques.
- Designed experiments varying redistribution regimes (taxation, charity) and studied wealth dynamics through inequality metrics and transaction data.
- Observed phase transitions in inequality dynamics near redistribution thresholds, illustrating non-linear macroeconomic effects.

Opinion Dynamics: De-polarisation of Social Opinions

September–October, 2024

Link to project: [GitHub](#)

- Built a computational model to study how social influence, echo chambers, and external “nudges” shape opinion polarization and convergence. Compared internal social influence and random dynamical nudges (RDNs) to assess their effects on depolarization and consensus formation.
- Implemented thorough parameter sweeps and visualized opinion distributions, demonstrating emergent social stability and fragmentation.

TEACHING EXPERIENCE

Tutor

September 2025 – Present

Faculty of Science, University of Amsterdam

Amsterdam, NL

- Tutoring students in introductory topics covered in first year of the **MSc Computational Science** program, such as SIR modeling and scientific computing basics.

Teaching Assistant

February 2025 – Present

Institute for Interdisciplinary Studies, University of Amsterdam

Amsterdam, NL

- Assisting in grading and teaching in **Complexity: Can it be Simplified?** course taught by Prof. Dr. Johan Bollen (2026) and Dr. Vítor Vasconcelos (2025).

Teaching Assistant

November 2024 – Present

Faculty of Social & Behavioural Sciences, University of Amsterdam

Amsterdam, NL

- Teaching Python programming, web-design (HTML, CSS, SQL and Flask), and algorithmic thinking to first-year **BSc Computational Social Science** students. Also responsible for general course content support (e.g. assignment, coursework and dataset creation).

Lecturer

May 2023 – June 2023

Erasmus School of Economics, Erasmus University Rotterdam

Rotterdam, NL

- Taught **Introduction to Python** to 75 students from non-technical background with a 97% pass rate, developed lectures and assignments for a 7-week course. Covered basics of OOP, Python syntax, Conditionals, Loops, Data Structures and important libraries such as Numpy, Pandas, Matplotlib and Seaborn.

Teaching Assistant

March 2023 – August 2023

Erasmus School of Economics, Erasmus University Rotterdam

Rotterdam, NL

- Developed and updated course materials for **Advanced Behavioral Economics** and **Economics of Negotiations**, aligning content with course objectives and learning outcomes.
- Created customized oTree (Python) games to provide practical experience in game theory principles and applications for students.

Teaching Assistant

October 2022 – November 2022

Erasmus University Rotterdam

Rotterdam, NL

- Guided 20 students in the **Python for Data Science** course, facilitating the successful completion of their final projects and evaluating their advancement in course topics

LEADERSHIP AND OUTREACH

Chairperson

September 2022 – August 2023

EFR×SABER, Erasmus University

Rotterdam, NL

- Led the board of the study association of the MSc Behavioural Economics program at Erasmus University Rotterdam, focused on academic, professional, and community engagement, organizing initiatives that connected students across backgrounds and supported an inclusive campus environment.

Head of Outreach and Business Development

September 2022 – August 2023

Turing Students Rotterdam, Erasmus University

Rotterdam, NL

- Tech-focused student association under Erasmus University. Activities included teaching bachelor-level courses on programming and data science, setting up collaborations with industry and academic stakeholders, and successfully acquiring monetary grants for various events.

President

September 2019 – August 2021

Association of Multicultural Computer Scientists (AMCS)

West Lafayette, IN, USA

- Led outreach and community-building initiatives for AMCS, coordinating student programming, mentoring members, and fostering academic engagement across campus while strengthening leadership, collaboration, and organizational growth.

Mentor

September 2020 – May 2022

Code Café, IEEE Purdue

West Lafayette, IN, USA

- Hosted informal coding sessions that fostered peer learning, collaboration, and an accessible technical community for students across experience levels.

Mentor

September 2020 – August 2021

StemPower Purdue

West Lafayette, IN, USA

- Mentored students in STEM through peer guidance, club activities, and academic support, helping create an encouraging community for students exploring technical fields.

Team Leader, International

March 2019 – September 2019

Boiler Gold Rush (BGR)

West Lafayette, IN, USA

- Led orientation groups for incoming international students, helping them navigate campus resources, build community, and transition confidently into life at Purdue.

AWARDS

Discovery Park Undergraduate Research Internship (DURI)

Awarded a scholarship to the Discovery Park Undergraduate Research Internship (DURI) program in Spring 2022.

[Link to project abstract.](#)

Dean's List, Semester Honors

Achieved 3.5+ GPA (semester and cumulative) in 2018, 2019 and 2021

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, HTML, CSS, JavaScript, C, C++, SOQL (Salesforce query language), Apex, L^AT_EX

Statistical Softwares: Stata, RStudio, SPSS, SAS